LUIS CALDERON

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Personal

Citizenship: U.S. (German in progress)
Languages: English, Spanish, German (B1)
Ethnicity: Latino

Placement

Director: Prof. Stephan Lauermann
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Coordinator: Dr. Silke Kinzig
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RESEARCH INTERESTS & METHODS

Macroeconomics, household heterogeneity, monetary and fiscal policy, Bayesian econometrics, functional data analysis

POSITION

University of Bonn *From August 2026*
Postdoctoral Researcher, Institute for Macroeconomics and Econometrics

EDUCATION

University of Bonn, Germany *Winter 2020 – Summer 2026*
Ph.D. in Economics, *summa cum laude*
M.Sc. in Economics with a specialization in research

University of Bonn, Germany *Winter 2018 – Summer 2020*
M.Sc. in Economics

Florida International University, Miami, FL *Fall 2013 – Fall 2017*
B.B.A. in Finance
B.A. in Economics

REFERENCES

Christian Bayer
University of Bonn
Institute for Macroeconomics
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Benjamin Born
University of Bonn
Institute for Macroeconomics
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Farzad Saidi
University of Bonn
Institute for Finance
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Moritz Kuhn
University of Mannheim
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JOB MARKET PAPER

Distributional Dynamics
With Christian Bayer and Moritz Kuhn. CEPR Discussion Paper 19829. Link.

We develop a new method for estimating high-frequency distributional dynamics across multiple economic variables. The method combines existing microdata sources that can differ in frequency, variable coverage, and measurement concepts. Central to the method is treating distributions as functional time series and representing their dynamics through a factor structure. We embed these factors in a state-space model and use Bayesian estimation to recover the underlying distributions from sparse and noisy observations. We apply the method to the joint distribution of consumption, income, and wealth, which is central to understanding how heterogeneous households respond to macroeconomic shocks. Using a wide range of U.S. microdata and macroeconomic time series, we construct the first quarterly estimates of this joint distribution from 1962 to 2024. We demonstrate that our approach reliably recovers its evolution, even when direct observations are sparse or unavailable. These new data provide previously unavailable empirical counterparts to the distributional dynamics at the heart of modern heterogeneous-agent macroeconomics.

WORKING PAPERS

Not All Oil Supply Shocks are Alike: The Macroeconomic and Distributional Impact of Oil Supply Shocks
May 2026. Link.

Coordination Is Overrated: The Assignment of Monetary and Fiscal Policy

Policy Rules and Policy Tools: The Price of Distributional Stabilization
October 2026. Link.

Spatial Standard Errors for Several Commonly Used M-estimators
With Leander Heldring. *R&R at the International Journal of Computational Economics and Econometrics.* Link.

WORKS IN PROGRESS

Distributional Dynamics: Germany
With Christian Bayer and Moritz Kuhn

Sectoral Job Levels and Productivity
With Christian Bayer, Moritz Kuhn and See-Yu Chan. Slides.

TEACHING EXPERIENCE

Bonn Graduate School of Economics

Class: Coding in Julia and Stata (PhD)	*Winter 2021–2025*
Class: Topics in Inequality (BSc)	*Summer 2024*
Class: Macroeconomics I (BSc)	*Summer 2025, 2026*
Winter School: BASEforHANK (for PhDs and central bankers)	*December 2025*

RESEARCH ASSISTANT EXPERIENCE

Department of Macroeconomics, University of Mannheim *January 2024 – September 2024*
PIs: Christian Bayer and Moritz Kuhn
Project: Datawork on Job Levels project

Institute of Macroeconomics and Econometrics, University of Bonn *June 2020 – 2022, 2025*
PIs: Christian Bayer, Benjamin Born, Ralph Luetticke, and Moritz Kuhn
Project 1: Shocks, Frictions, and Inequality in US Business Cycles
Project 2: BASEforHANK
Master Thesis: The Effects of Income Uncertainty on Liquidity, Default, and Prosperity

Institute on Behavior & Inequality (briq) *May 2019 – June 2020*
PI: Prof. Leander Heldring
Project 1: The Cost of State-building: Evidence from Germany
Project 2: Spatial standard errors for several commonly used M-estimators

MEMBERSHIPS AND ROLES

2024 – present: Member of Collaborative Research Center Transregio 224 "Economic Perspectives on Societal Challenges"
2022 – 2023: Member of RTG 2281 "The Macroeconomics of Inequality"
2021 – 2024: BGSE Student Body Representative
2024 – 2025: Kreisliga Basketball (PG), Volunteer for U18 games
2022 – 2026: Bonner Liga Basketball (PG and Treasurer)
2021 – 2026: NRW Liga Chess Player

ABSTRACTS

Working Papers

Not All Oil Supply Shocks are Alike: The Macroeconomic and Distributional Impact of Oil Supply Shocks. Link.

Oil supply shocks shape U.S. inequality, and *the type of* oil supply shock matters for who bears the cost. In a Bayesian VAR that embeds the joint distribution of U.S. household income, consumption, and wealth alongside a standard oil-market block, sudden production shortfalls (Baumeister and Hamilton, 2019) raise inequality, whereas oil supply news (Känzig, 2021) mostly reduces it. These differences are not visible in the aggregate dynamics, which appear similar across shocks. A causal mediation analysis (Dufour and Wang, 2024) of these aggregate dynamics shows these shocks operate through distinct channels and it is this mechanism wedge, rather than the aggregate footprint, that determines who bears the cost. The analysis shares the view of Kilian and Lewis (2011) that policy responses should depend on the underlying causes of oil price shocks.

Coordination Is Overrated: The Assignment of Monetary and Fiscal Policy

After an oil supply shock, how should the central bank and the fiscal authority coordinate? I study this question using a Bayesian VAR that tracks macroeconomic aggregates and the joint distribution of household consumption, income, and wealth, with identified monetary and fiscal policy shocks providing the levers for constructing policy counterfactuals. Letting the two authorities play simultaneous and sequential games under different assignments of inflation, unemployment and consumption inequality, I find coordination matters less than assignment and timing. A central bank that targets only inflation more than doubles the loss relative to a cooperative planner. A central bank with a dual mandate that moves first, with fiscal policy responsible for consumption inequality, comes closer to the cooperative outcome. Leaving consumption inequality unassigned is costly: even a cooperative planner with only a dual mandate does worse than every non-cooperative arrangement in which the central bank keeps its dual mandate and leads.

Policy Rules and Policy Tools: The Price of Distributional Stabilization. Link.

This paper examines how much monetary and fiscal policy can stabilize consumption inequality after an oil supply shock, and at what aggregate cost. I model policy instruments jointly with the distribution of U.S. household consumption, income, and wealth, and compare the outcomes policy can achieve under alternative mandates. The shock raises inflation, depresses real activity, and compresses the consumption distribution. Under single mandates, the policy rate has limited capacity to stabilize oil-induced inflation but substantial capacity to stabilize the consumption tails (the 90/10 spread). A dual mandate over inflation and unemployment mainly stabilizes real activity, with gains partly extending to the untargeted consumption and income spreads. Adding even a small weight on the consumption spread to the dual mandate removes much of the remaining spread loss at a modest aggregate cost, borne mostly through unemployment; stabilizing the spread further comes at a convex cost. Fiscal instruments extend how far consumption inequality can be stabilized, but do not lower its price. Finally, stabilizing the tails does not stabilize the whole distribution: a target on all ten consumption deciles relative to aggregate consumption is harder to meet with the policy rate, but benefits more from fiscal policy.

Spatial Standard Errors for Several Commonly Used M-estimators (with Leander Heldring). Link.

We provide a unified implementation of existing asymptotic theory that computes Conley-style spatial HAC standard errors for a wide range of commonly used (non-linear) estimators. We cover OLS, logit, probit, Poisson, and negative binomial regressions, as well as the fixed-effects estimators *areg* and *reghdfe*—extending commonly used publicly available routines (currently limited to linear models) to nonlinear M-estimators and fixed-effects workflows. We provide Stata and Python software implementing the procedure.

Works in Progress

Sectoral Job Levels and Productivity (with Christian Bayer, Moritz Kuhn and See-Yu Chan). Slides.

Bayer and Kuhn (2023) show that a worker's *job level* explains nearly half of the differences in pay between individual workers in Germany. Job levels (*Leistungsgruppen*) are an official five-step ladder on which the German statistical office classifies jobs by their complexity, the autonomy they allow, and the responsibility they carry. We build the long-run, economy-wide counterpart: a newly digitized quarterly panel of West German wages and employment by job level, 1958–2021, merged with sector productivity accounts (EU-KLEMS). We report three findings. **First**, part of the productivity growth that standard growth accounting attributes to technology is in fact workers moving into higher job levels: about 10% of measured total factor productivity (TFP) growth in our sectors over 1975–2006, and 30% in Finance,

19% in Mining and 10% in Manufacturing. About half of it comes from the growth of the highest level we observe and most of the rest from the shrinking of the two lowest levels. **Second**, across sectors, the average job level explains 57% of the differences in average pay within a year, and it predicts output per hour worked but not output per unit of capital. **Third**, about twice as much of women's real wage growth as men's came from moving into higher job levels, among both blue- and white-collar workers. Beneath all three, about 86% of real wage growth over 1975–2006 came from higher pay *within* job levels and about 14% from workers moving *up* (25% in 1995–2006). The same split appears in the separate survey that replaced ours in 2007, so the conclusions do not depend on linking the two surveys.

ADDITIONAL INFORMATION

Seminars & Conferences	**2026:** Annual Conference of the Central Bank of Brazil (Banco Central do Brasil); North American Summer Meeting of the Econometric Society (June); CESifo Area Conference on Macro with Micro Data (June, Poster); ICMAIF (University of Crete, May) **2025:** CASFI Summer Meeting; BCFM; CRC TR 224 Retreat **2024:** Summer Meeting of the SED (Barcelona, Presenter); Banca d'Italia (Rome, Poster); Money Conference (Board of Governors); Center for Advanced Studies in Finance and Inequality (Bonn); Firm Heterogeneity and Macroeconomics (Bonn/Mannheim, 2022–2024) **2023:** HANK Conference (Berlin)
Informal Supervision of Master Theses	**2025:** Marius Trovik (1.0) **2024:** Zsigmond Szajbely (1.0) **2023:** Luke Liscio (1.0) **2021:** Anna Sanchez (1.7)
Schools & Workshops	**Summer Schools:** Macroeconomics and Financial Markets (June 2022, Dean Corbae), DSE 2024 **Spring Schools:** Macroeconomics and Big Data (April 2024, Domenico Giannone, Giorgio Primiceri) **Workshops:** Empirical Methods in Macroeconomics (May 2024, Christian Wolf)
Personal	**Citizenship:** U.S. (German in progress) **Ethnicity:** Latino **Languages:** English, Spanish, German (B1)